

NATIONAL INCOME

National Income Accounting (NIA)

- NIA is the measurement of indicators of national output/income; .e.g. GDP.
- GDP (Gross Domestic Product) is one of the most important concepts in economics. It measures the total market value of output of a nation. It is the total money value of goods and services produced in an economy during a given period of time, usually a year.

Definition of GDP:

- The market value of good i (V_i) is equal to $P_i \cdot Q_i$
- GDP = sum of the market values of all **final** goods and services produced within the year.
- $GDP = P_1 \times Q_1 + P_2 \times Q_2 + P_3 \times Q_3 + \dots + P_n \times Q_n$

$$GDP = \sum_{i=1}^n V_i = \sum_{i=1}^n P_i \cdot Q_i$$

(Cont.)

- Final goods and services - goods and services that are not purchased for the purpose of producing other goods and services or for resale, e.g. bread is a *final* product but flour is an *intermediate* product.
- Including both intermediate goods and final goods in the calculation of GDP will result in “double counting”.

(Cont.)

- Why is GDP (total production) an important number?
 - The more goods and services a country produces, the more **POTENTIAL** ability it has to satisfy needs.
 - The more goods and services a country produces, the more total income (**national income**) it generates and the better its **POTENTIAL** ability to satisfy needs.

Real vs. Nominal GDP:

- Why is GDP measured in pounds (dollars, euros, ... etc.)?
 - We measure GDP in pounds because we must know the *value* of how much we produced.
 - We can not add up wheat, cattle, health care ... etc.
 - We must use pounds as a “**common denominator**” to be able to add values of different goods and services.
 - We add **pounds** worth of wheat + **pounds** worth of cattle + ... etc.

GDP at current
prices

Nominal GDP

- We multiply quantities purchased by the price at which they were sold on that year.

Example: For 2009 GDP, we use prices paid in 2009 .

- Because we use prices of **2009** to compute GDP for **2009** , we say that we used “**current**” prices.

GDP

The sum of expenditures on all goods and services produced over the year

Year	Good X		Good Y			GDP
2009	Price	Quantity	Price	Quantity	PriceXQuantity	
	\$1	100	\$3	50	$(1 \times 100) + (3 \times 50) =$	250

If prices rise...

Quantities produced
are the same in both
years!

The use of prices
distort our view
of production

But GDP Increased!

Good Y		GDP	
Price	Quantity	PriceXQuantity	
\$3	50	$(1 \times 100) + (3 \times 50) =$	250
Price	Quantity	PriceXQuantity	
\$2	100	$(2 \times 100) + (6 \times 50) =$	500

Real GDP:

- We **have** to use prices to measure GDP.
- But we must **eliminate the effect** of changing prices.
- To eliminate the effect of changing prices, we **pretend** that prices did not change.

(Cont.)

Year	Price X	Quantity X	Price Y	Quantity Y	\$worthX	\$worthY	Nominal GDP
					$P_x * Q_x$	$P_y * Q_y$	
2008	1	100	2	50	1×100	$+ 2 \times 50$	$= 200$
2009	2	150	3	100	2×150	$+ 3 \times 100$	$= 600$
2010	3	200	4	150	3×200	$+ 4 \times 150$	$= 1200$

Year	Price X	Quantity X	Price Y	Quantity Y	\$worthX	\$worthY	Nominal GDP
					$P_x * Q_x$	$P_y * Q_y$	
2008	1	100	2	50	1×100	$+ 2 \times 50$	$= 200$
2009	1	150	2	100	1 $\times 150$	$+ 2 \times 100$	$= 350$
2010	1	200	2	150	1 $\times 200$	$+ 2 \times 150$	$= 500$

Real GDP: Use **Constant** Prices

Year	Price X	Quantity X	Price Y	Quantity Y	\$worthX	\$worthY	Real GDP
	in base year		in base year		$P_x * Q_x$	$P_y * Q_y$	
2008(base)	1	100	2	50	1×100	$+ 2 \times 50$	$= 200$
Use base year prices to compute REAL GDP							
2009	1	150	2	100	1×150	$+ 2 \times 100$	$= 350$
2010	1	200	2	150	1×200	$+ 2 \times 150$	$= 500$

GDP at constant
prices

Real GDP

- For **Real** GDP, we multiply quantities purchased by the price at which they were sold on **the base year**.

Example: For 2009 GDP, we use prices prevailing in **base year**.

- Because we use prices of **base year** to compute GDP for **2009**, we say that we used “**constant**” prices.

Potential GDP:

- It is the real GDP that the economy produces when **all** workers and factories (all resources) are fully employed.

Economic Growth:

- Increases in GDP come from:
 - Increases in the number of workers (labor resources) and/or
 - Increases in the number of factories (capital stock) and/or
 - Improvements in technology

(Cont.)

The growth rate of potential GDP depends on:

- The growth rate of the labor force
- The growth rate of the nation's capital stock
- The rate of technical progress

The Circular Flow of Income:



(Cont.)

- Hypothetical two-sector economy (households and businesses – consumers and producers/investors).
- Only final goods included.
- GDP = flow of final products (flow of product or expenditure approach) = flow of costs (earnings or cost or income or value-added approach).
- Profit is a residual cost. It enters the lower loop as factor payment.

(Cont.)

■ Expenditure (product) approach to GDP:

$GDP = \text{Consumption} + \text{Investment}$ (2 sectors)

$GDP = \text{Consumption} + \text{Investment} + \text{Government exp. on goods and services}$ (3 sectors)

$GDP = \text{Consumption} + \text{Investment} + \text{Government exp. on goods and services} + (\text{Exports} - \text{Imports})$ (4 sectors)

$GDP = C + I + G + (X - M)$

■ Earnings (income/value-added) approach to GDP:

$GDP = \text{Wages} + \text{Interest} + \text{Rents} + \text{Profits} + \text{Indirect taxes}$

■ **The expenditure approach and the earnings (value-added) approach give **identical** results.**

Readings: Samuelson & Nordhaus, Ch. 21